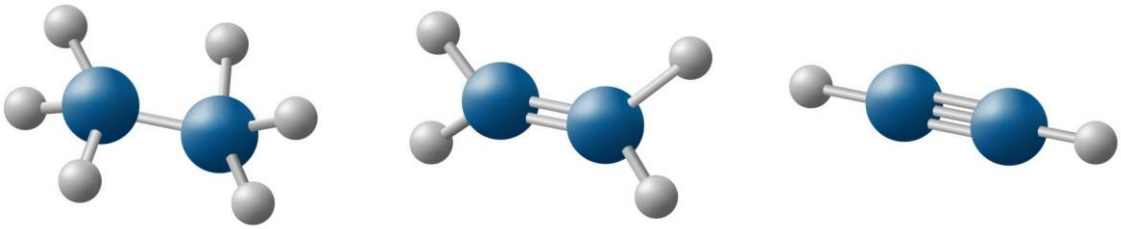
⍰ HYDROCARBONS

Introduction

Hydrocarbons are organic compounds consisting only of carbon and hydrogen atoms. They form the basis of organic chemistry and are classified into aliphatic and aromatic hydrocarbons. Understanding their structure, properties, and reactions is essential for NEET, as this chapter forms the foundation for advanced organic chemistry.

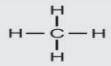
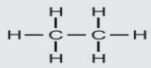
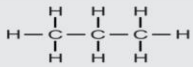
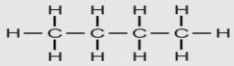
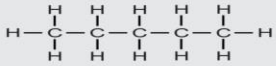
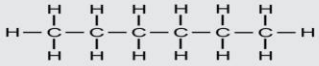
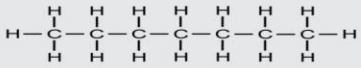
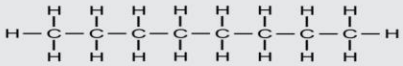
Classification of Hydrocarbons

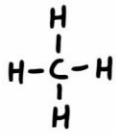
Hydrocarbons are broadly classified into alkanes, alkenes, alkynes, and aromatic hydrocarbons. Alkanes are saturated hydrocarbons with single bonds, while alkenes and alkynes are unsaturated with double and triple bonds respectively. Aromatic hydrocarbons contain benzene rings.



Alkane Alkene Alkyne

HYDROCARBONS (Alkanes)

	METHANE
	ETHANE
	PROPANE
	BUTANE
	PENTANE
	HEXANE
	HEPTANE
	OCTANE



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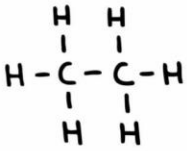

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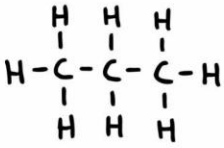

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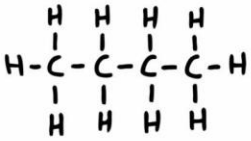

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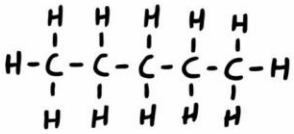

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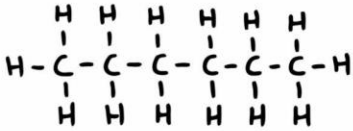

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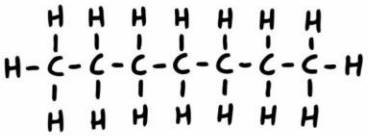

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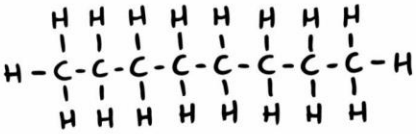
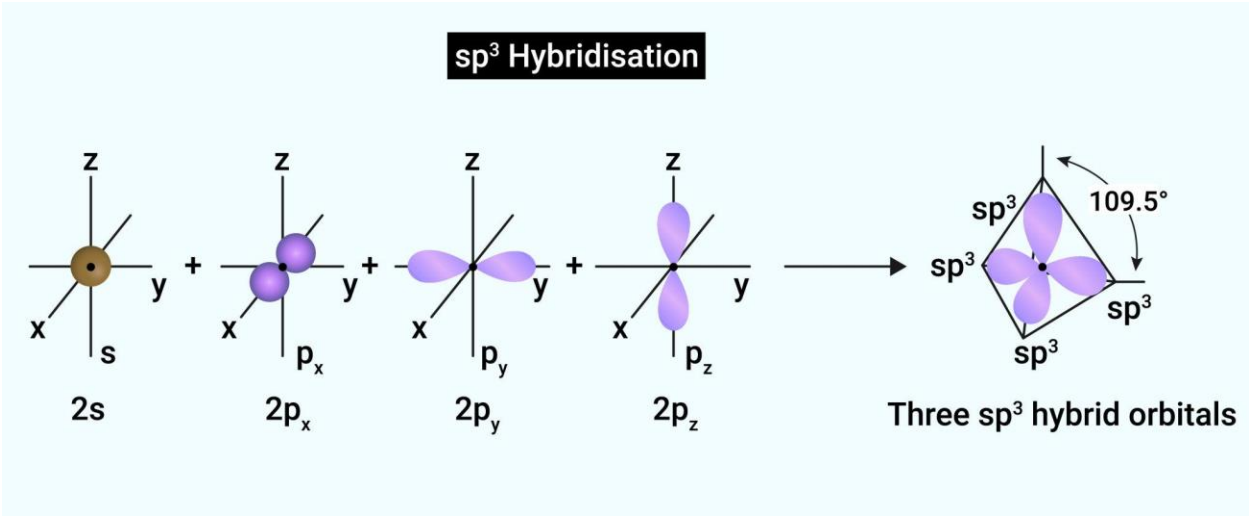


OCTANE

Alkanes

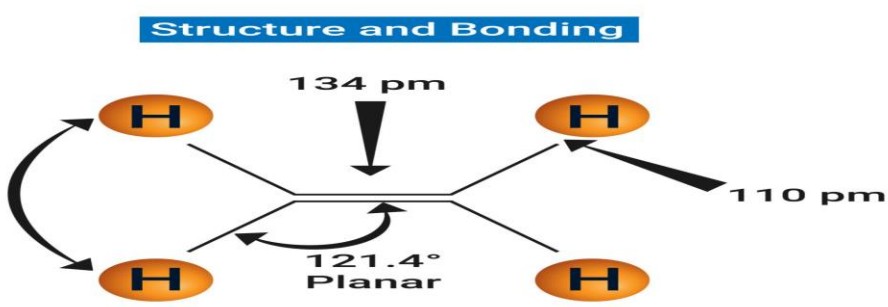
Alkanes are saturated hydrocarbons with general formula C_nH_{2n+2}. They have only single bonds and exhibit sp^3 hybridization with tetrahedral geometry.

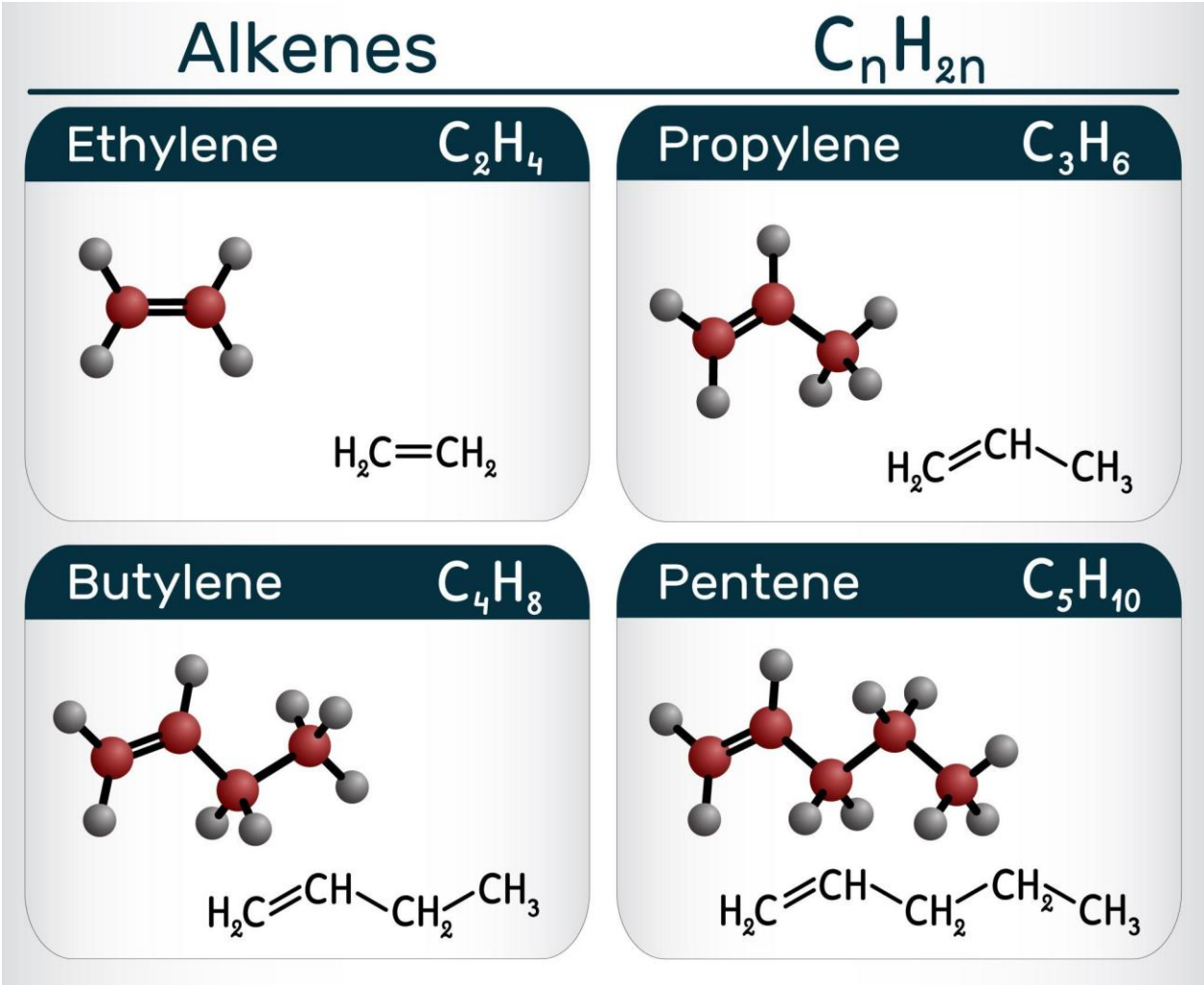


They undergo substitution reactions and are relatively less reactive.

Alkenes

Alkenes contain at least one double bond and have the general formula C_nH_{2n}. They exhibit sp^2 hybridization and planar structure.

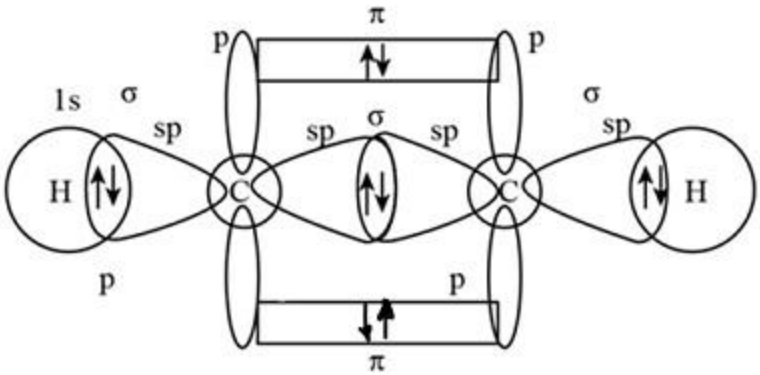


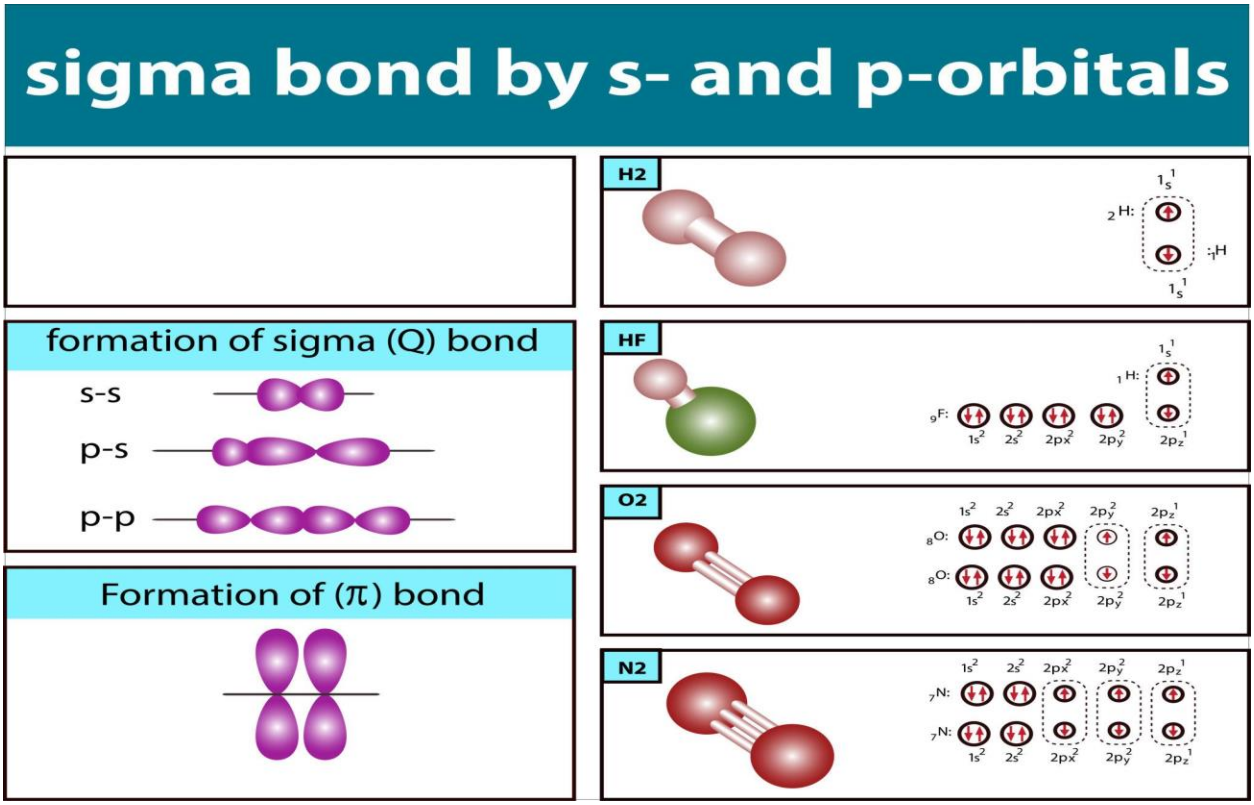


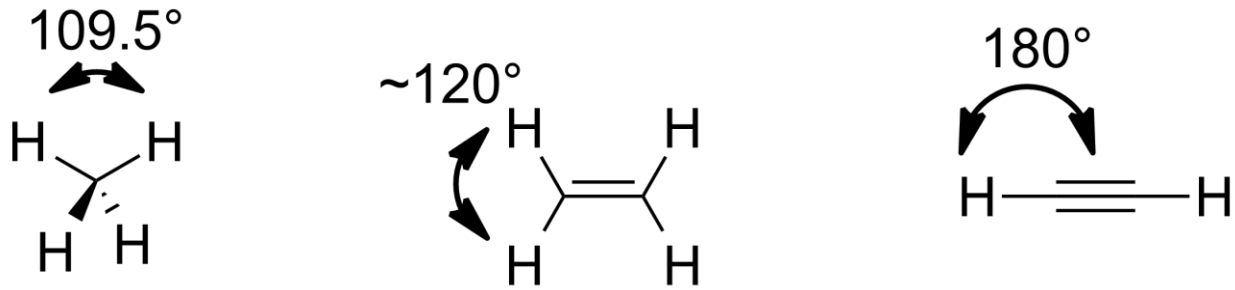
They undergo addition reactions due to the presence of a double bond.

Alkynes

Alkynes contain at least one triple bond and have the general formula C_nH_{2n-2}. They exhibit sp hybridization and linear geometry.



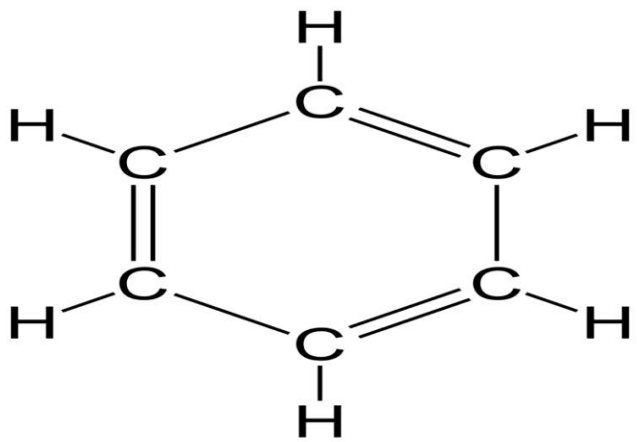




They are more reactive than alkenes and participate in addition reactions.

Aromatic Hydrocarbons

Aromatic hydrocarbons contain benzene rings with delocalized π electrons. They follow Huckel's rule and exhibit resonance.



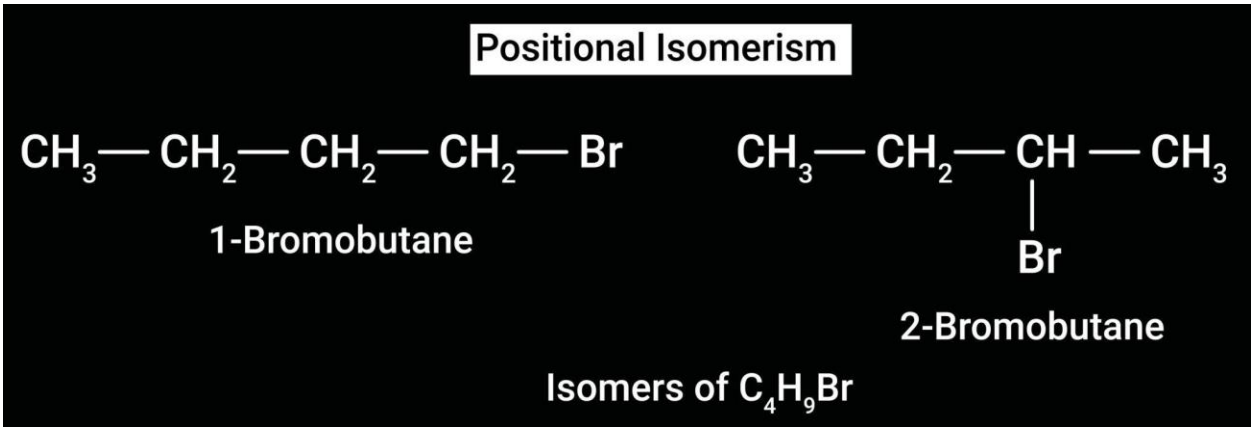
Benzene

C_6H_6

These compounds are stable and undergo substitution reactions.

Isomerism

Isomerism occurs when compounds have the same molecular formula but different structures. Structural and geometrical isomerism are important types in hydrocarbons.



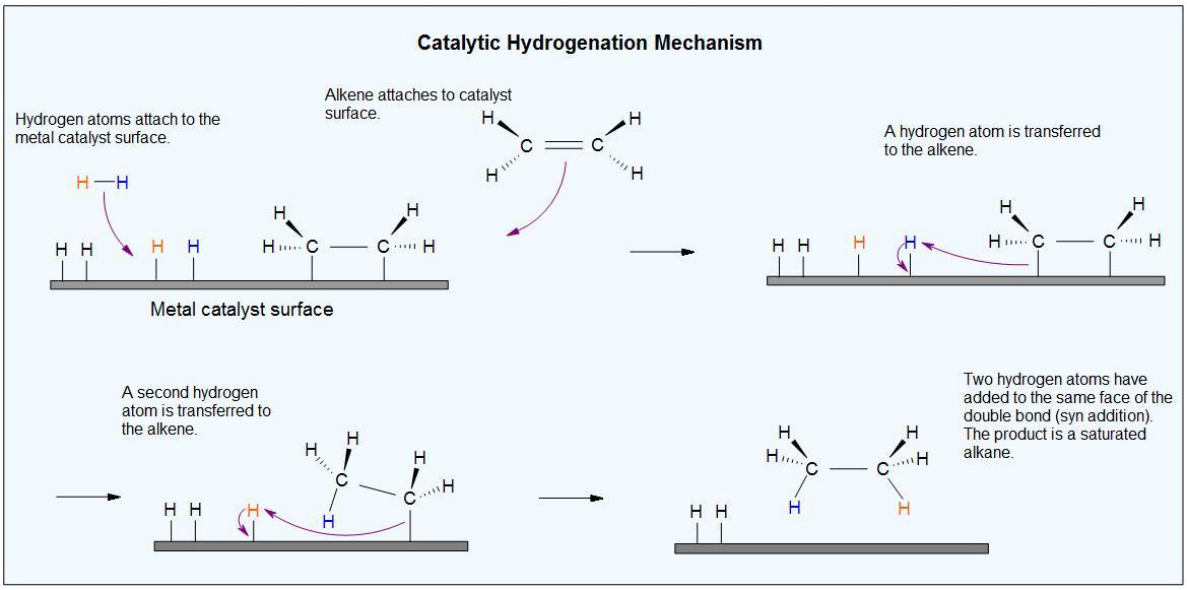
Isomerism affects physical and chemical properties.

Preparation of Hydrocarbons

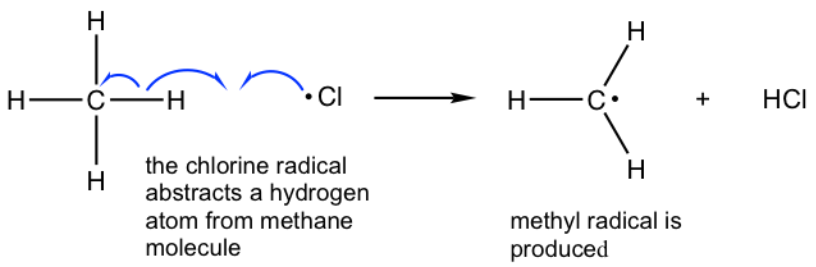
Hydrocarbons can be prepared by various methods such as hydrogenation, elimination reactions, and decarboxylation. These methods help in synthesizing different types of hydrocarbons.

Chemical Reactions

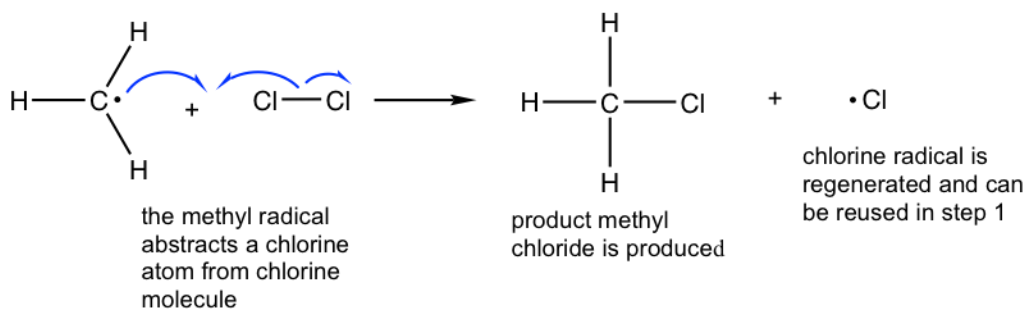
Hydrocarbons undergo different reactions based on their structure. Alkanes undergo substitution reactions, while alkenes and alkynes undergo addition reactions. Aromatic compounds undergo electrophilic substitution.

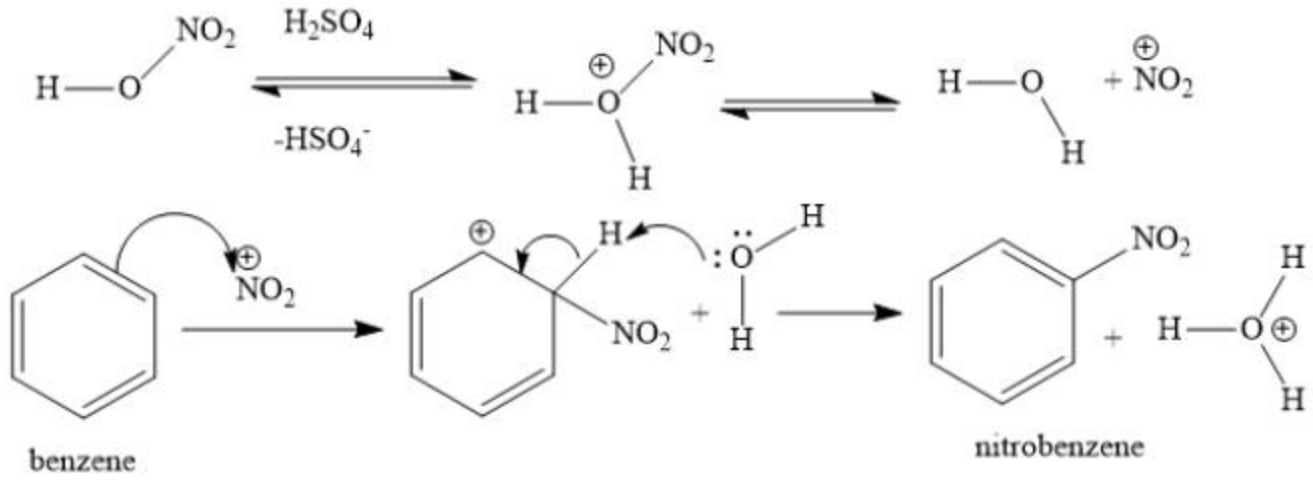

Catalytic Hydrogenation Mechanism
Hydrogen atoms attach to the metal catalyst surface.
Alkene attaches to catalyst surface.
A hydrogen atom is transferred to the alkene.
Metal catalyst surface
A second hydrogen atom is transferred to the alkene.
Two hydrogen atoms have added to the same face of the double bond (syn addition). The product is a saturated alkane.

step 1:


the chlorine radical abstracts a hydrogen atom from methane molecule
methyl radical is produced
+ HCl

step 2:


the methyl radical abstracts a chlorine atom from chlorine molecule
product methyl chloride is produced
chlorine radical is regenerated and can be reused in step 1



Understanding these reactions is essential for solving problems.

Environmental Aspects

Hydrocarbons are major components of fossil fuels. Their combustion releases energy but also contributes to pollution and global warming.